

DP024

Physics 2

Semester 2

Session 2022/2023

2 hours



UNIT FIZIK
KOLEJ MATRIKULASI KELANTAN

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME PRE-EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
NO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

NAME: _____

PRAC: _____

MATRIC NO: _____

Kertas soalan ini mengandungi 5 halaman bercetak
This question paper consists of 5 printed pages.

Answer **all** questions.
Jawab **semua** soalan.

1. (a) A progressive wave is represented by the following equation: $y = 10 \sin (4 \pi t + x)$ where y is in centimeters and t is in seconds. Determine,
- (i) the direction of wave propagation.
 - (ii) the wavelength.
 - (iii) frequency of the wave.
 - (iv) vibrational velocity at time $t = 0$ s, for the particle is at $x = 5.0$ cm.

- (b) The equation of a stationary wave along a stretched string is given as

$$y = 3 \cos \left(\frac{\pi}{4} x \right) \sin (60 \pi t)$$

Where y and x are in cm and t is in seconds. Find

- (i) the velocity of the progressive waves.
- (ii) the separation between two adjacent nodes.

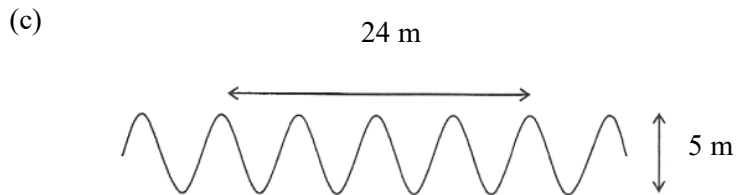
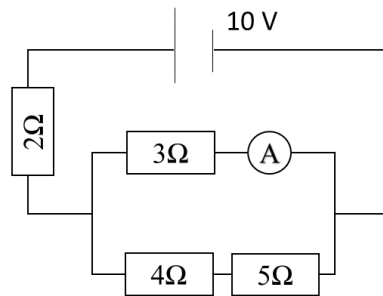


FIGURE 1

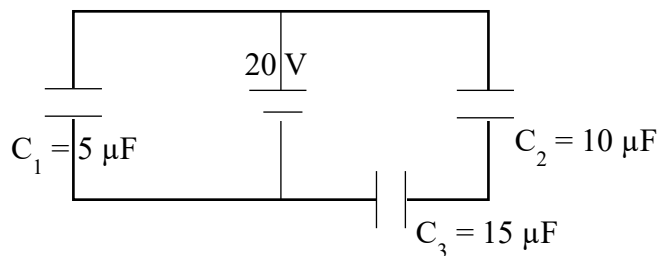
FIGURE 1 shows a progressive wave traveling at a speed of 30 m s^{-1} to the right. Write the equation of the progressive wave.

2.

**FIGURE 2**

- (a) **FIGURE 2** shows a circuit with combination of four resistors connected to a 10 V battery. The ammeter shows a reading of 1.76 A. Determine
- the potential difference across resistor 3Ω .
 - the current provided by the battery.
 - the potential difference in resistor 4Ω .
- (b) The resistivity of a copper wire is $1.72 \times 10^{-8} \Omega \text{ m}$. If the wire has a diameter of 1.00 mm and a length of 50 m, calculate the resistance of the wire.

3.

**FIGURE 3**

- (a) **FIGURE 3** shows three capacitors connected to a 20 V battery.
- Determine the total capacitance of the circuit.
 - Calculate the charge stores in C_3 .
- (b) A capacitor of $100 \mu\text{F}$ is connected in series to a resistor $200 \text{ k}\Omega$ and a 6 V battery. Calculate the current flow through the resistor in time, $t = 5 \text{ s}$.

4.

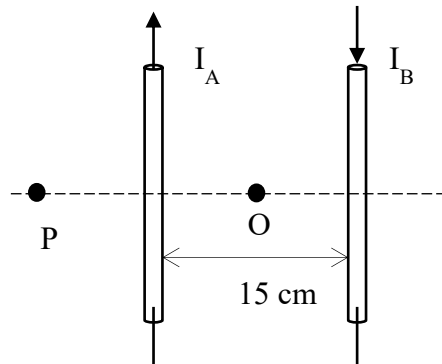


FIGURE 4

(a) **FIGURE 4** above shows two parallel long wires **A** and **B** which carries current, $I_A = 10 \text{ A}$ and $I_B = 20 \text{ A}$ respectively in opposite directions. The wires are separated 15 cm apart. The resultant magnetic field is zero at point **P** along the line connecting both wires. Determine,

- (i) The value of resultant magnetic at point **O** which is midway between the wires.
- (ii) The distance of point **P** from wire **A**.

(b)

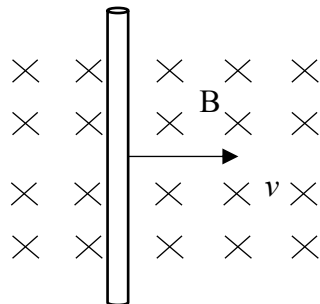


FIGURE 5

FIGURE 5 shows a copper rod of length 25 cm moves to the right with velocity 20 ms^{-1} in a uniform magnetic field, $B = 0.5 \text{ T}$. The resistance of the copper rod is 2Ω .

- (i) Calculate the induced current in the copper rod.
- (ii) Show the direction of the induced current in the figure given.

5. (a) An object is placed 5.0 cm from a converging mirror with a curvature radius of 20.0 cm.
- (i) Sketch a ray diagram to show the formation of the image.
 - (ii) Find the position of the image.
- (b) A convex lens of focal length 10 cm, is used to form a real image **TWO** times larger than an object. If an image height is 2.5 cm and formed behind the lens, determine;
- (i) Object should be placed.
 - (ii) Image distance.
 - (iii) The height of the object.
 - (iv) State the characteristics of the image.
6. (a) In Young's double-slit experiment, the distance between the slit and the screen is 1.5 m and the separation of the double slits used is 0.2 mm. If the width between the **TWO** fourth dark fringes is 4.0 cm, determine;
- (i) Wavelength of light used in the experiment.
 - (ii) The fourth bright from the center bright.
 - (iii) Fringe separation between consecutive bright.
- (b) A beam of light with a wavelength of 600 nm is illuminated on a double-slit producing an interferences pattern on the screen. Another wavelength with λ is also illuminated on a double-slit and produces another interferences pattern on the screen. If third order dark fringe from 600 nm wavelength coincides with the fourth-order bright wavelength λ , determine the value of λ .

END OF QUESTION PAPER
KERTAS SOALAN TAMAT